

Scripting 101

From Command Line Basics to Data Analysis Automation

 bash — ~

```
$ ./start_course.sh --from zero --to reproducible-workflows  
Ready. Type, don't paste.
```

By the end of the course you will be able to

1 Work entirely from the Linux command line in WSL

2 Write robust Bash scripts that process thousands of files

3 Create reproducible software environments with Conda

4 Analyse and visualise data with Python and pandas

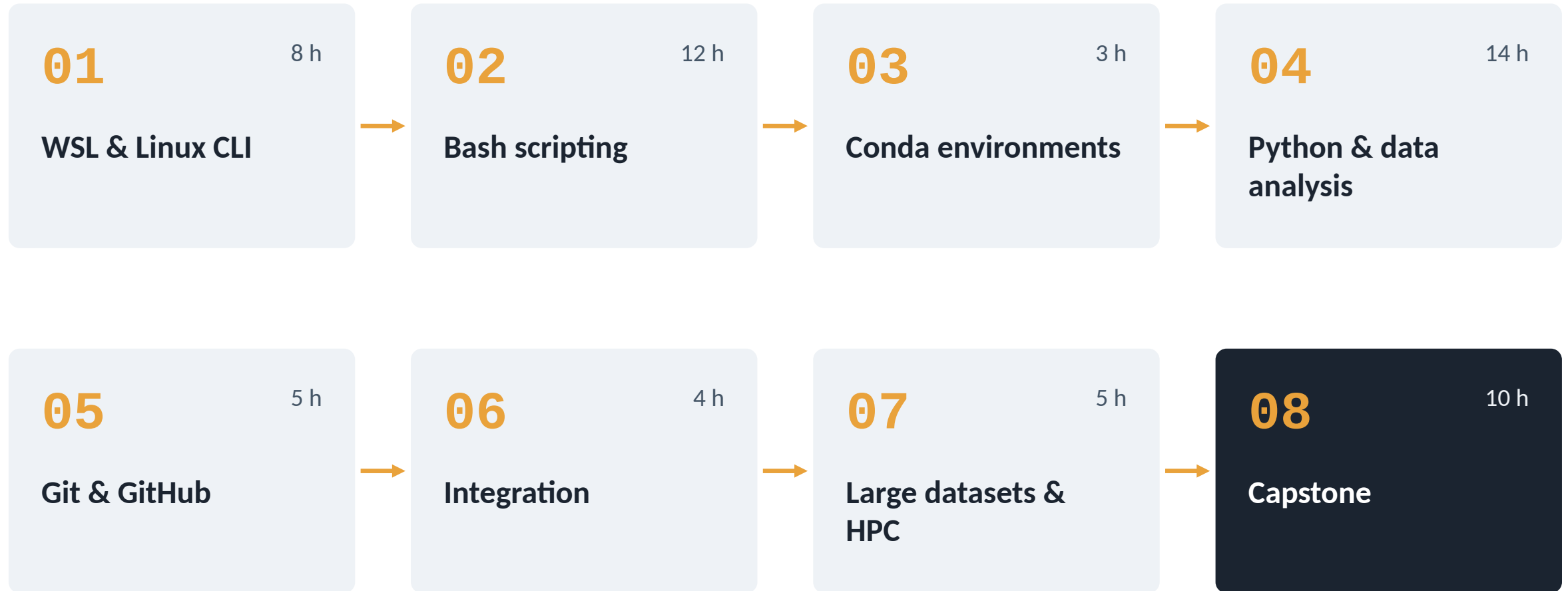
5 Turn exploratory analyses into tested command-line programs

6 Version-control projects with Git and publish them on GitHub

7 Process datasets larger than memory, and verify data integrity

8 Deliver a one-command, reproducible analysis that others can rerun

Eight modules, one progression



Beginner → fundamental skills → applied skills → intermediate → advanced → independent workflow development

How we work in this course



Type, don't paste

Typing builds memory, and fixing your own typos teaches you to read errors.



Read the error message

The last line usually says exactly what is wrong. Read it out loud.



Build step by step

Run each piece and look at the output before adding the next one.



Ask after 15 minutes

Stuck? Ask a neighbour, then an instructor. Bring the exact error text.



Scripts, not clicks

If you did it twice, it belongs in a script. Scripts are repeatable.



Explain every line

Use any reference you like, but you must be able to explain your code.

Our data: the HARVEST maize trial

- **Field trial:** 6 sites × 5 years × 4 varieties × 4 nitrogen rates × 3 reps
- **Weather:** 30 daily files, one per site-year
- **Sensors:** 168 soil-moisture logger files in nested folders
- **Genomics:** FASTA, FASTQ.gz, VCF.gz, expression counts
- **Realistic problems** hidden inside: duplicates, typos, missing values, broken files. You will find them.

```
~/scripting-101/datasets
```

```
datasets/  
├── field_trials/  
│   ├── plot_yields.csv      1,443 plots  
│   └── sites.csv  
├── weather/  
│   └── weather_AMES_2020.csv ... (30)  
├── sensors/AMES/2024-07-01/  
│   └── logger_01.csv ... (168)  
├── genomics/  
│   ├── reference.fasta  
│   ├── reads/S01.fastq.gz ... (8)  
│   ├── variants.vcf.gz  
│   └── expression_counts.tsv  
└── messy/    CRLF, spaces in names
```

Setup check: is everything ready?

```
bash
# in Ubuntu (WSL)
$ cd ~/scripting-101
$ conda activate scripting101
$ python --version
Python 3.12.14
$ seqkit version
seqkit v2.x
$ git --version
git version 2.53.0
$ bash scripts/bash/checksums.sh verify datasets
All files OK
```

conda activate

switches to the course software environment (Module 3)

checksums.sh verify

proves your copy of the data is identical to everyone else's (Module 7)